

Ollama + AMD GPU Setup & Validation Notes

Overview

This document contains the commands used to:

- Verify Ollama installation and API availability
 - Verify AMD GPU and ROCm support
 - Download Large Language Models (LLMs)
 - Run models locally
 - Automate the entire setup process
-

Terminal History Summary

Terminal 1 – Environment Validation

Check Ollama API

```
curl http://127.0.0.1:11434/api/tags
```

Purpose:

- Verifies Ollama server is running.
 - Lists all locally available models.
-

Check Running Models

```
ollama ps
```

Purpose:

- Displays currently loaded models.
 - Shows GPU/CPU memory usage.
-

Check AMD GPU

```
lspci | grep -i amd
```

Purpose:

- Confirms Linux detects the AMD GPU.
-

Check ROCm Device

```
ls /dev/kfd
```

Purpose:

- Verifies ROCm kernel driver is loaded.

Expected Output:

```
/dev/kfd
```

Check ROCm Status

```
rocm-smi
```

Purpose:

- Displays GPU health.
 - Shows VRAM utilization.
 - Confirms ROCm installation.
-

Terminal 2 – Model Downloads

Download DeepSeek-R1 70B

```
ollama pull deepseek-r1:70b
```

Purpose:

- Downloads DeepSeek-R1 70B model.
-

Verify Models

```
curl http://127.0.0.1:11434/api/tags
```

Purpose:

- Confirms DeepSeek model exists locally.
-

Download Llama 3.1 70B

```
ollama pull llama3.1:70b
```

Purpose:

- Downloads Llama 3.1 70B model.
-

Verify Models Again

```
curl http://127.0.0.1:11434/api/tags
```

Purpose:

- Confirms all required models exist.
-

Run DeepSeek

```
ollama run deepseek-r1:70b
```

Purpose:

- Starts DeepSeek-R1 70B for inference.
-

Terminal 3 – Run Llama

Start Llama 3.1

```
ollama run llama3.1:70b
```

Purpose:

- Starts Llama 3.1 70B for inference.
-

Complete Workflow

Step 1 – Verify Ollama

```
curl http://127.0.0.1:11434/api/tags
```

Step 2 – Verify AMD GPU

```
lspci | grep -i amd
```

Step 3 – Verify ROCm Driver

```
ls /dev/kfd
```

Step 4 – Verify ROCm Installation

```
rocm-smi
```

Step 5 – Download Models

```
ollama pull deepseek-r1:70b  
  
ollama pull llama3.1:70b
```

Step 6 – Verify Models

```
curl http://127.0.0.1:11434/api/tags
```

Step 7 – Run Models

DeepSeek

```
ollama run deepseek-r1:70b
```

Llama

```
ollama run llama3.1:70b
```

Automation Script

Save as:

```
automation.sh
```

Contents:

```
#!/bin/bash  
  
echo "===== "  
echo "Checking Ollama Server"  
echo "===== "  
curl http://127.0.0.1:11434/api/tags
```

```

echo
echo "=====
echo "Checking AMD GPU"
echo "=====
lspci | grep -i amd

echo
echo "=====
echo "Checking ROCm KFD Device"
echo "=====
ls /dev/kfd

echo
echo "=====
echo "Checking ROCm Status"
echo "=====
rocm-smi

echo
echo "=====
echo "Pulling DeepSeek-R1 70B"
echo "=====
ollama pull deepseek-r1:70b

echo
echo "=====
echo "Pulling Llama 3.1 70B"
echo "=====
ollama pull llama3.1:70b

echo
echo "=====
echo "Installed Models"
echo "=====
curl http://127.0.0.1:11434/api/tags

echo
echo "=====
echo "Loaded Models"
echo "=====
ollama ps

echo
echo "=====
echo "Environment Validation Complete"
echo "=====

```

Make Script Executable

```
chmod +x automation.sh
```

Run:

```
./automation.sh
```

Full Setup Script (Starts Ollama Automatically)

Save as:

```
setup_ollama_amd.sh
```

Contents:

```
#!/bin/bash

pkill ollama 2>/dev/null

nohup ollama serve > ollama.log 2>&1 &

sleep 10

curl http://127.0.0.1:11434/api/tags

lspci | grep -i amd

ls /dev/kfd

rocm-smi

ollama pull deepseek-r1:70b

ollama pull llama3.1:70b

ollama ps
```

Make executable:

```
chmod +x setup_ollama_amd.sh
```

Run:

```
./setup_ollama_amd.sh
```

Useful Troubleshooting

Ollama Server Not Running

Error:

```
could not connect to ollama server
```

Fix:

```
ollama serve
```

Or:

```
nohup ollama serve > ollama.log 2>&1 &
```

Verify Ollama Process

```
ps aux | grep ollama
```

Verify Loaded Models

```
ollama ps
```

List Downloaded Models

```
ollama list
```

or

```
curl http://127.0.0.1:11434/api/tags
```

Environment Checklist

- ☐ Ollama Installed
 - ☐ Ollama Server Running
 - ☐ AMD GPU Detected
 - ☐ ROCm Installed
 - ☐ /dev/kfd Available
 - ☐ DeepSeek-R1 70B Downloaded
 - ☐ Llama 3.1 70B Downloaded
 - ☐ Model Loads Successfully
 - ☐ GPU Utilization Visible in rocm-smi
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Last Updated: June 2026 Purpose: AMD GPU + Ollama Local LLM Deployment & Validation Guide